UNIVERSITY OF PISA

Lesson 7

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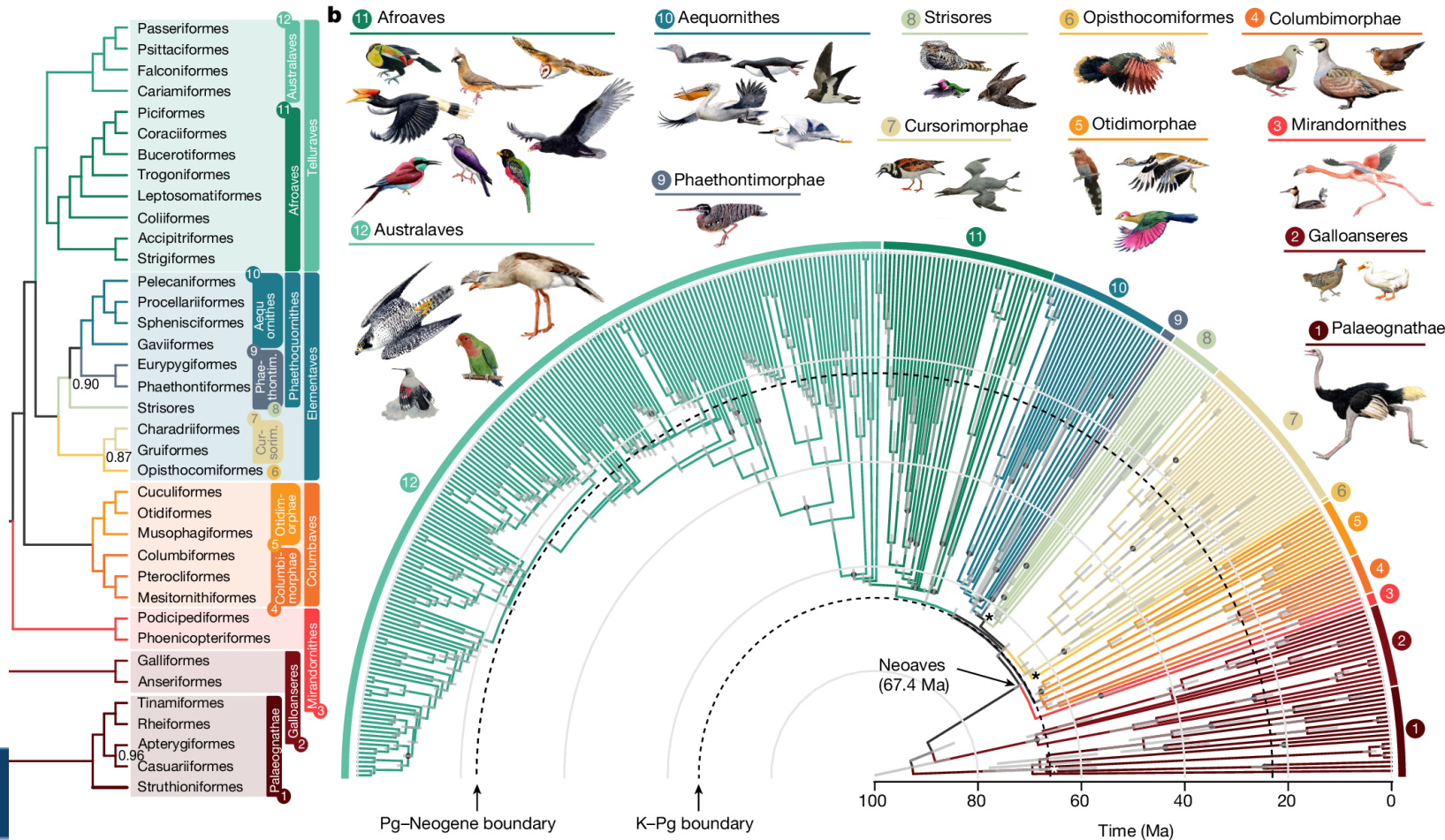
PHYLOGENETIC ANALYSIS

Phylogenetics

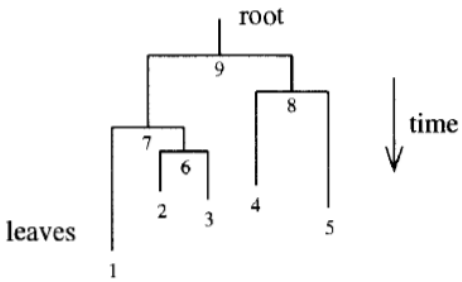
- Studies the evolutionary history and relationships among individuals or spices.
- The Phylogenetic tree represents this information

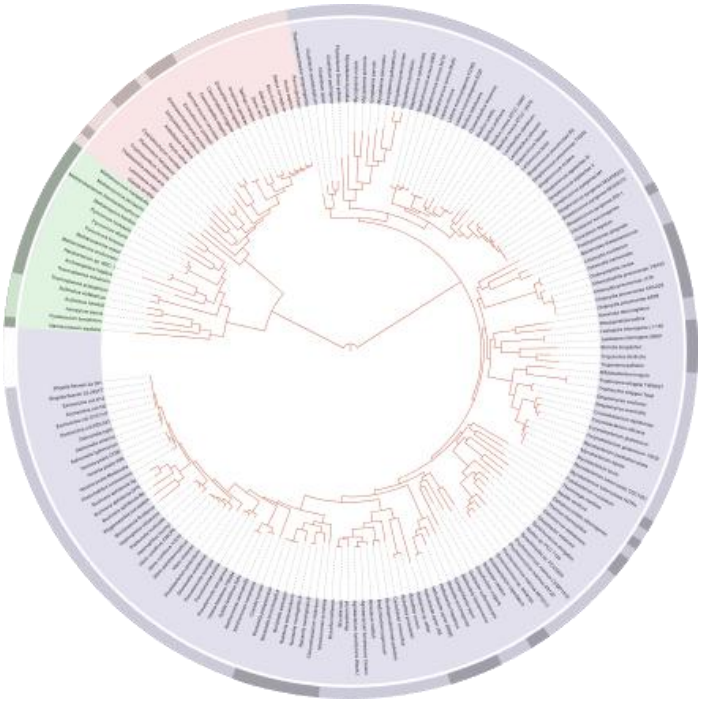


Stiller, J., Feng, S., Chowdhury, AA. *et al.* Complexity of avian evolution revealed by family-level genomes. *Nature* **629**, 851–860 (2024). https://doi.org/10.1038/s41586-024-07323-1

Phylogeny and phylogenetic trees

- Molecular similarities suggest that organisms are related by common ancestors
- The relationships are intrinsically hierarchical
  - leaves are organisms or sequences
  - internal nodes are common ancestors
  - the height of the edges is proportional to the evolutionary distance



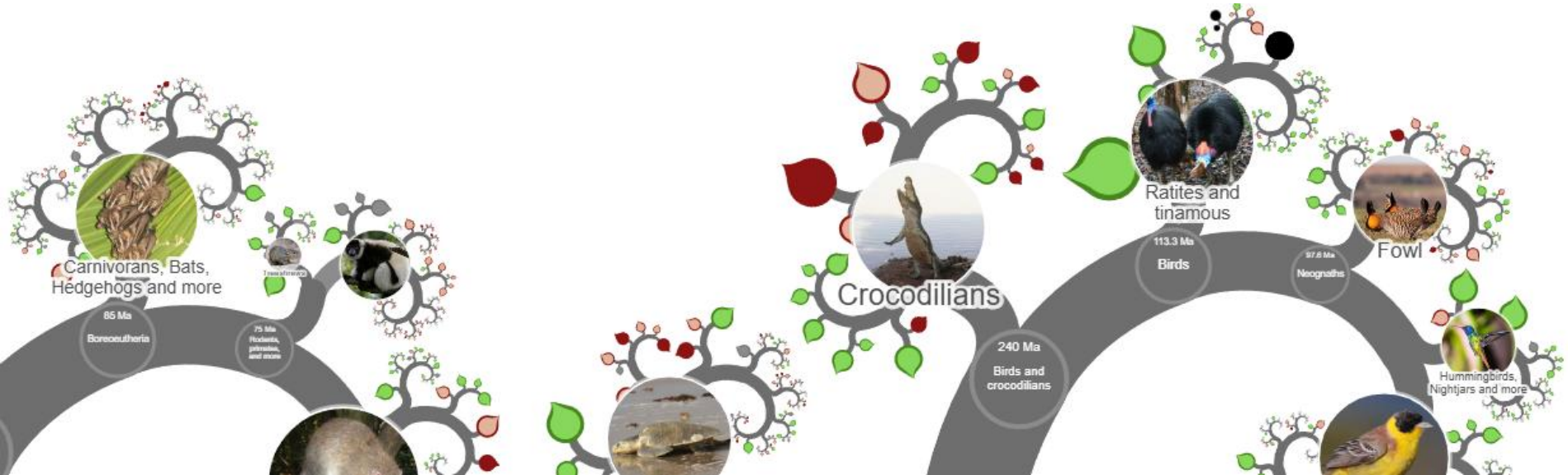


Tree of life from Wikipedia

https://www.onezoom.org/

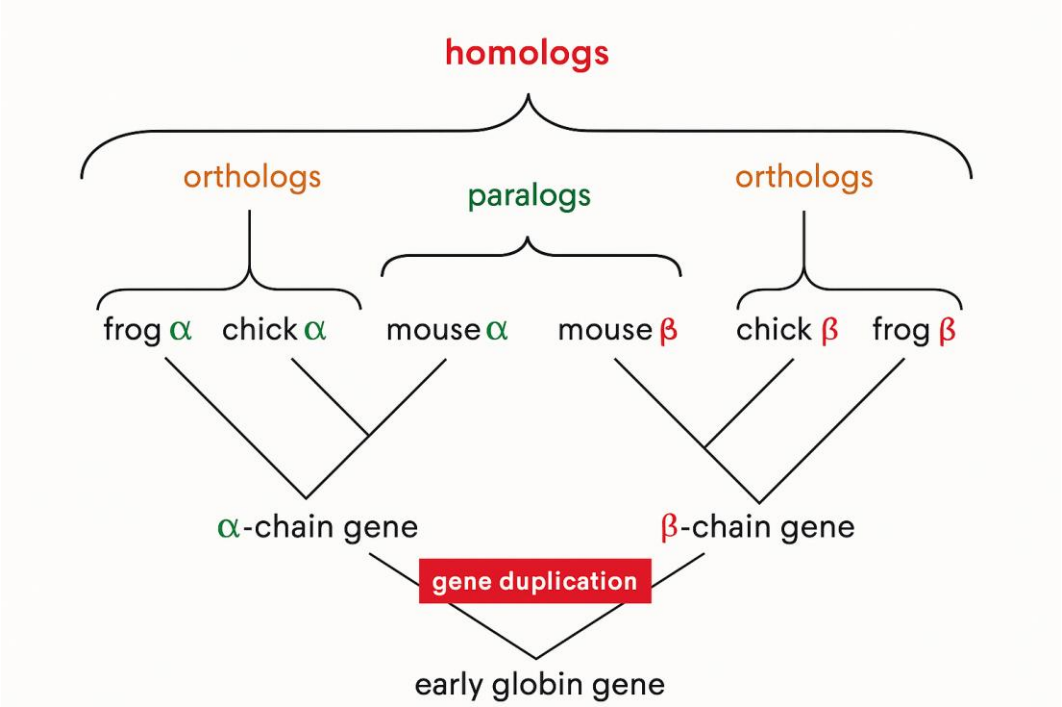
Trees can contain different entities

- Interspecific phylogenesis
  - On leaves there are taxa or spices
  - Typically inferred from conserved genomic regions (as RNAs)



Phylogeny and phylogenetic trees

- Building phylogenetic trees
  - Traditionally using morphological characteristics
  - More recently - molecular characteristics - common ancestor genes



- ***Orthologues:*** genes that diverged by speciation . Two different 'versions' of a common ancestor gene resulted in different species
  - To be used to infer relation between species
- ***Paralogues:*** genes are duplicated in the same species and then each evolves to have a different function
  - Can be used to build a phylogeny of paralogues (even within the same species)

https://biodataprog.github.io/GEN220_2019/Bioinformatics/02_orthology.html

Building phylogenetic trees

- It is an optimization problem:
  - We need to find the best possible tree given an input of related sequences assumed that they have a common ancestor
  - Given a set of sequences there are many different possible trees
  - Difficult: when *n* increases the number of possible solution become quickly huge
  - How can we define a suitable objective function?

objective functions

Three main classes of algorithms:

1. **Distance based algorithms** - convert sequence differences into pairwise distances and build the tree (e.g., Neighbor-Joining, Minimum Evolution) that best fits those distances
2. **Maximum parsimony** - choose the tree that explains the data with the fewest total evolutionary changes across characters (less mutations needed).
3. **Statistical/Bayesian methods** - use explicit models of sequence evolution to estimate trees (maximum likelihood finds the tree with highest likelihood, while Bayesian inference samples the posterior over trees to quantify uncertainty).

1. Building phylogenetic trees from sequences using distances

Align sequences (MSA).

Compute the **distance** matrix.

Build the tree

Define **objective function** and evaluate the tree

1. Distances between sequences

- Many possible definitions
- A simple one: calculate the percentage of gaps or mismatches
  - Consider f_{ij} the fraction of residues that are different after aligning two sequences s_i and s_j

S_i	A	A	C	G	C	G	T	A	T	T	G	C	C	G	G
S_j	A	A	C	G	A	G	G	A	T	A	C	C	G	G	G

$$f_{i,j} = 5/15 = 0.33$$

1. Distances between sequences

A more complex one: **Jukes-Cantor distance**

$$d_{ij} = -3/4\ln(1 - 4/3p)$$

With p the observed fraction of differing sites between two aligned sequences (Hamming distance)

d = JC-corrected evolutionary distance

- Four nucleotides with equal base frequencies (¼ each).
- All substitutions have the same rate (A↔C↔G↔T symmetric).
- Sites evolve independently under a continuous-time Markov process.

1. The objective function

- It can be defined as an error function
- Minimize the differences between the distances in the tree and the distances in the matrix using, for example, the sum of square errors:

$$score(T) = \sum_{i,j \in S} \left(d_{i,j}(T) - D_{i,j}\right)^2$$

With *d* the distances in the tree and D the distances in the distance matrix

- Since the space of all possible trees given a *n* number of sequences is quite big, and it grows very rapidly the problem of finding the best tree is NP-hard

Many approaches: heuristic algorithms

1. Building the tree

- Apply hierarchical clustering algorithms to build the tree
  - bottom-up (agglomerative) methods:
    - UPGMA - Unweighted Pair Group using Arithmetic Average
    - Neighbour-Joining
  - the distance between clusters will define the length of the edges

1.1-UPGMA

- **Idea:** Build the tree starting from the bottom and going up, by successively joining the closest clusters
- **Steps:**
  - Assign each sequence to its own cluster C_i
  - Compute the distance between all cluster pairs as the pairwise distance between sequences
  - Place one tree leaf for each sequence at height 0
  - While number of clusters >1
    - Select the closest clusters i and j and join them into one cluster k
    - Add a parent node to i and j at height $d_{ij}/2$
    - Compute the distance between the new cluster and the rest as the average of the pairwise distance of all sequences

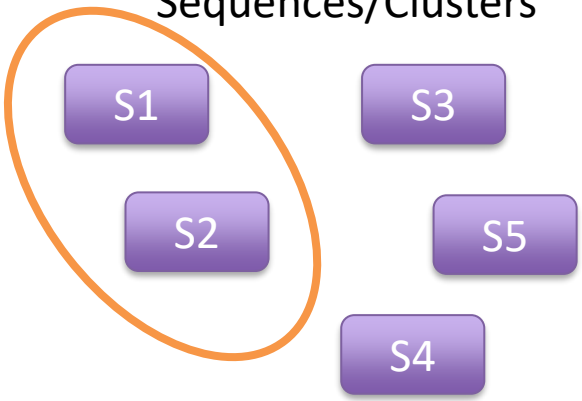
UPGMA example

Distance matrix

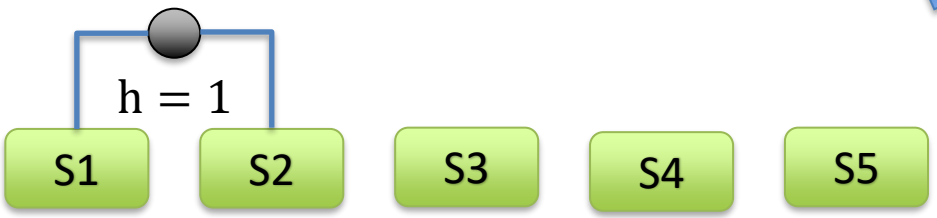
	S1	S2	S3	S4
S2	**2**			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

mean

Sequences/Clusters



Tree

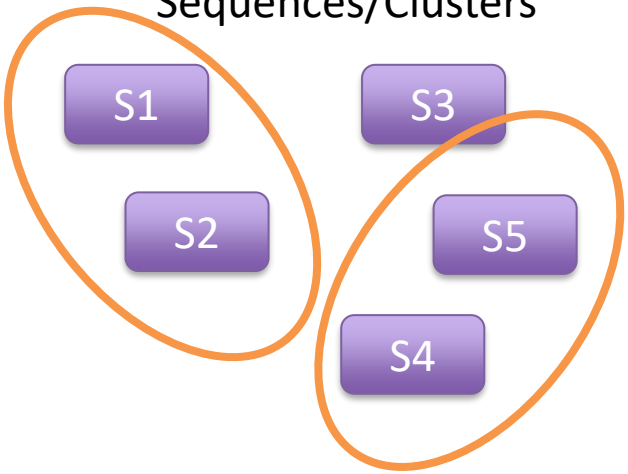


UPGMA example

Distance matrix

	S1,S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	**3**

Sequences/Clusters

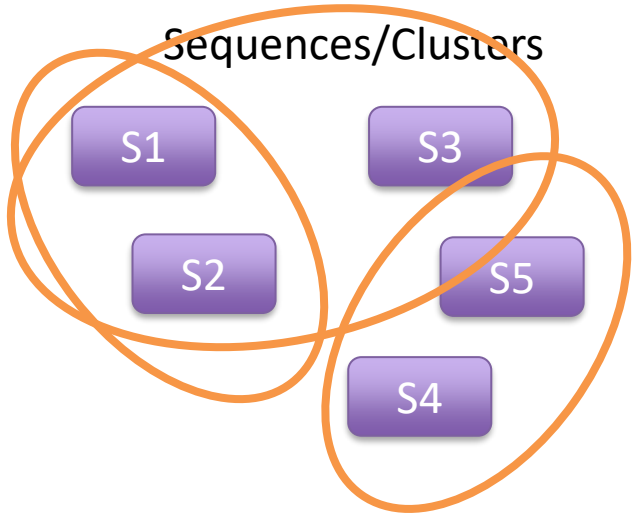


Tree

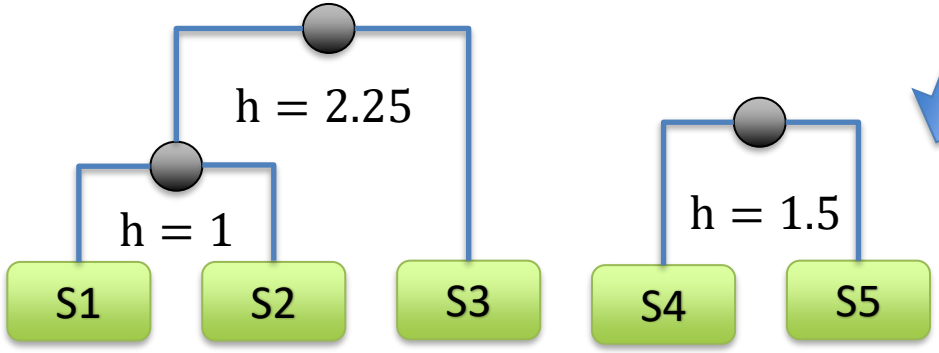
UPGMA example

Distance matrix

	S1,S2	S3
S3	4.5	
S4,S5	7.25	5



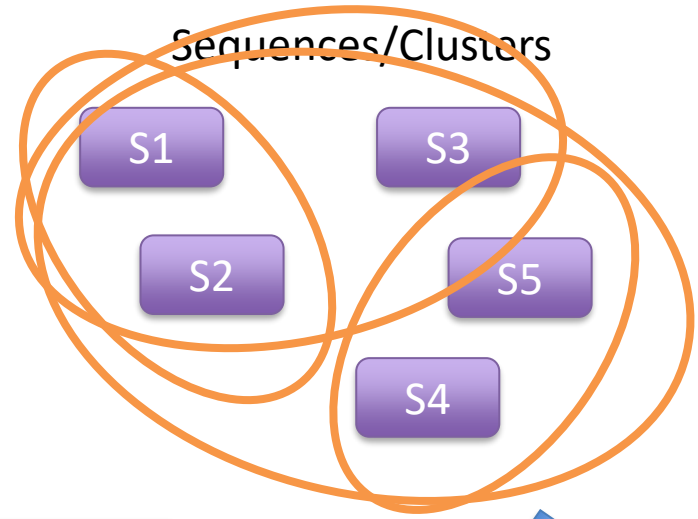
Tree



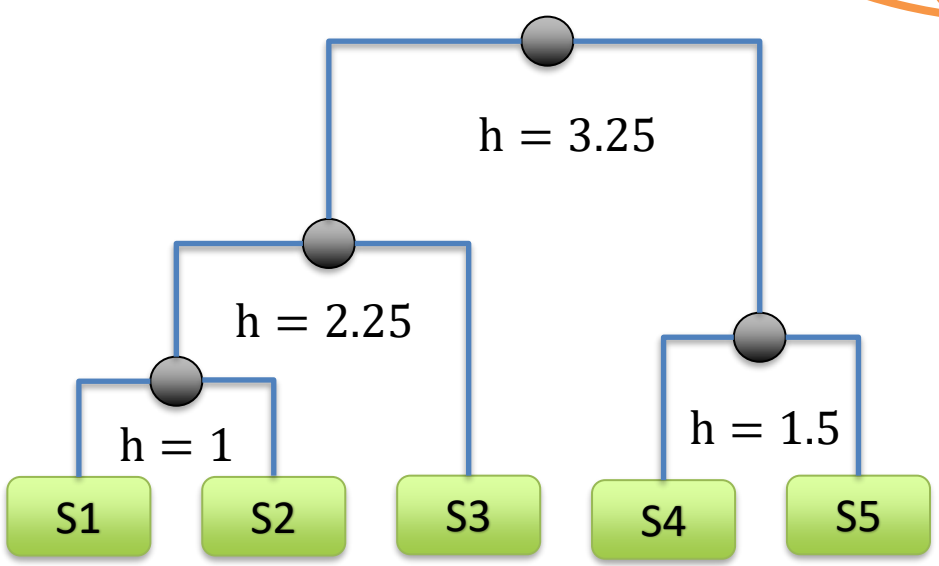
UPGMA example

Distance matrix

	S1,S2, S3
S4,S5	6.5

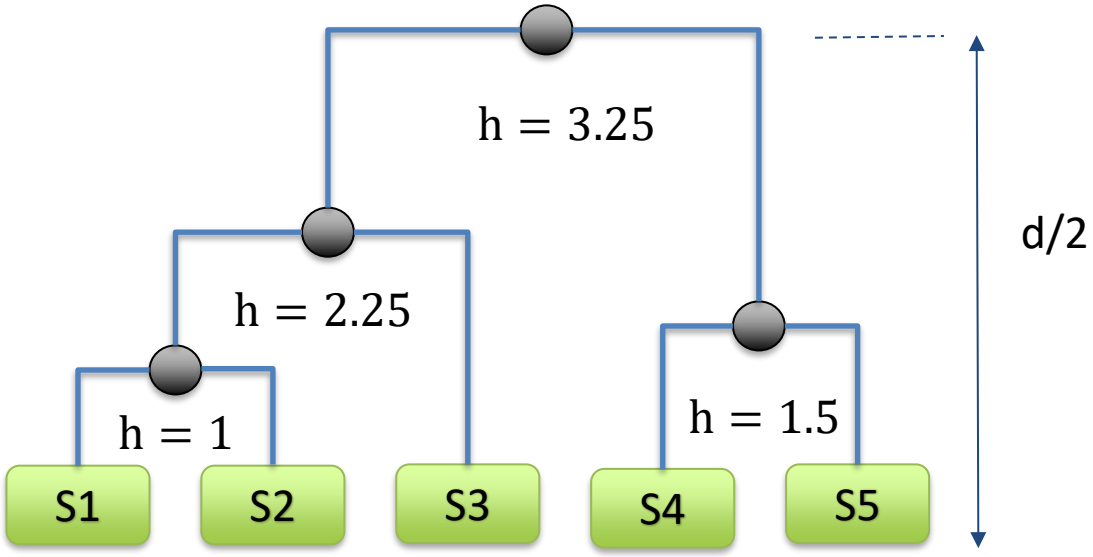


Tree



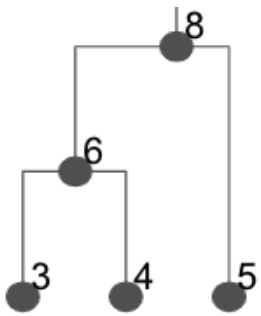
UPGMA

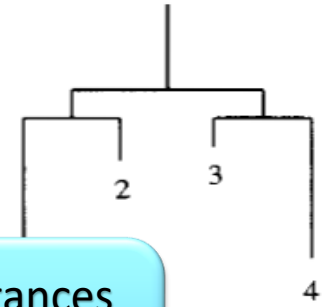
- Are edge lengths well defined?
  - Yes, because distances increase as the algorithm proceeds
    - we always select the cluster pair with smallest distance



UPGMA

- Results in a tree where all leaves are on the same level
- Works if distances are ultrametric - otherwise the tree will be incorrect
  - **trees where every tip is the same distance from the root**, so branch lengths represent **time** and the tree satisfies the ultrametric inequality for any three taxa.
  - For any 3 sequences s_i, s_j, s_k the distances d_{ij}, d_{ik}, d_{jk} are either all equal, or two are equal and the third is smaller
- All lineages have accumulated equal total time since their common origin → a (strict) **molecular clock** holds. Internal node heights are divergence times; the distance between two tips equals





It als ssumes a cons e…

often false!

Non ultrametric distances => **Neighbour-Joining algorithm**

1.2-Neighbour joining

- Idea: shortest pairwise distance is not enough to identify two neighbours but we also seek to select cluster pairs where the nodes are apart from the others.
- Starting from D we define a new distance Q_{ij} by subtracting from D_{ij} the average distance of s_i and s_j with all other leaves

$$Q_{i,j} = (n-2)D_{i,j} - \sum_{k=1}^{n} D_{i,\mathrm{k}} - \sum_{k=1}^{n} D_{j,\mathrm{k}}$$

- Minimum Q_{ij} identifies true neighbours

1.2-Neighbour joining: step

- In each step the D matrix is updated
- If *u* is the new cluster (new node in the tree) crated by the join of nodes *a* and *b*, the distances are calculated as:

$$D_{ui} = 1/2(D_{ai} + D_{bi} - D_{ab})$$

- From this a new Q is calculated